

Customer Shopping Behavior Analysis

Data Analytics Project

Technologies – Python, PostgreSQL, SQL, Power BI

Summary

This project analyzes customer shopping behavior using a dataset containing 3,900 purchase records across multiple product categories, demographic groups, and purchasing preferences.

The analysis was designed to identify patterns in customer spending, product preferences, discount usage, subscription behavior, customer loyalty, shipping preferences, and age-group revenue contribution.

Python was used for data preparation and exploratory analysis, PostgreSQL was used for structured SQL-based business analysis, and Power BI was used to develop an interactive dashboard for communicating the findings.

The analysis indicates that customer behavior varies across demographic and transactional characteristics. A substantial proportion of transactions are associated with returning or loyal customers, while non-subscribed customers represent the majority of transactions.

Productlevel differences in ratings and discount usage also provide opportunities for targeted marketing and promotional planning.

The overall purpose of the project is not simply to describe historical data, but to translate customer behavior into **actionable business questions and recommendations**.

1. Introduction

Retail businesses generate large volumes of customer transaction data. However, raw transactional data has limited value until it is transformed into information that decisionmakers can understand and act upon.

Customer behavior analysis helps businesses answer questions such as:

- Who are the customers?
- What products do they purchase?
- How much do they spend?
- How frequently do they purchase?
- Which products receive better reviews?
- How do discounts influence purchasing behavior?
- Are repeat customers more engaged?
- How does subscription status relate to spending?

This project addresses these questions using a structured data analytics workflow.

2. Problem Statement

Management wants to understand customer purchasing behavior in order to improve sales, customer satisfaction, and long-term loyalty.

The organization is particularly interested in differences across demographics, product categories, discounts, reviews, seasons, payment preferences, shipping methods, and customer purchasing history.

Main Business Question

How can customer shopping data be leveraged to identify behavioral trends, improve customer engagement, and optimize marketing and product strategies?

3. Project Objectives

The project has the following objectives:

Objective 1

Clean and prepare the raw customer dataset for analysis.

Objective 2

Explore customer demographics and purchasing behavior.

Objective 3

Analyze revenue and purchasing patterns using SQL.

Objective 4

Identify customer segments based on purchasing history.

Objective 5

Analyze the relationship between discounts and purchasing behavior.

Objective 6

Compare subscribed and non-subscribed customers.

Objective 7

Identify highly rated and frequently purchased products.

Objective 8

Create an interactive Power BI dashboard.

Objective 9

Translate analytical findings into business recommendations.

4. Dataset Description

The dataset contains:

- 3,900 records
- 18 variables
- 25 unique products
- 4 product categories
- 50 locations

The dataset includes customer demographics, purchase information, product information, shipping preferences, payment methods, reviews, discounts, subscription status, and purchase history.

5. Data Preparation

Python was used as the primary data-preparation environment. **Step**

1: Data Loading

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	1
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	1
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN

The CSV dataset was loaded using Pandas.

Step 2: Data Inspection

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

The dataset was examined using:

```
df.info()  
df.describe()  
df.head()
```

This helped identify:

- Data types
- Missing values
- Numerical distributions
- Categorical variables
- Dataset dimensions

Step 3: Missing Values

The `Review Rating` column contained 37 missing values.

These values were imputed using the median review rating of the corresponding product category.

Step 4: Standardization

Column names were standardized using snake_case naming.

Step 5: Feature Engineering

Additional analytical features were created, including:

- Age groups
- Purchase-frequency measures

Step 6: Database Loading

The cleaned dataset was transferred to PostgreSQL for SQL analysis.

6. Exploratory Data Analysis

The exploratory analysis focused on:

Demographics

- Age
- Gender
- Location

Product Behavior

- Category
- Product
- Size
- Color

- Season

Transaction Behavior

- Purchase amount
- Discount usage
- Shipping method
- Payment method

Customer Engagement

- Subscription status
- Previous purchases
- Purchase frequency
- Review rating

7. SQL Business Analysis

SQL was used to convert raw records into business-level metrics.

The major analyses included:

1. Revenue by gender

	gender 	revenue 
1	Female	75191
2	Male	157890

2. High-spending discount users

	customer_id 	purchase_amount 
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
Total rows: 839		Query complete 00:00:00

3. Top-rated products.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. Shipping comparison

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. Subscription analysis

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. Discount-dependent products

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

7. Customer segmentation

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. Top products by category

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. Repeat buyers and subscriptions

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. Revenue by age group

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

8. Key Findings

8.1 Revenue Distribution

Male customers generated \$157,890 in recorded purchase revenue compared with \$75,191 from female customers.

This difference should be considered in the context of the dataset's customer composition and should not automatically be interpreted as a difference in individual customer value.

8.2 Subscription Behavior

There are:

- 1,053 subscribed transactions
- 2,847 non-subscribed transactions

Average purchase values are approximately:

- Subscribers: \$59.49
- Non-subscribers: \$59.87

The difference in average transaction value is small, suggesting that subscription status alone is not strongly associated with transaction value in this dataset.

8.3 Customer Loyalty

The customer segmentation analysis identifies:

- 83 New
- 701 Returning
- 3,116 Loyal

This indicates that the dataset contains a large share of customers with established purchase histories.

8.4 Product Ratings

The highest average ratings identified were:

- Gloves: 3.86
- Sandals: 3.84
- Boots: 3.82
- Hat: 3.80
- Skirt: 3.78

8.5 Shipping

Average purchase value:

- Standard: \$58.46
- Express: \$60.48

This indicates a modest difference between the two shipping groups.

8.6 Discount Usage

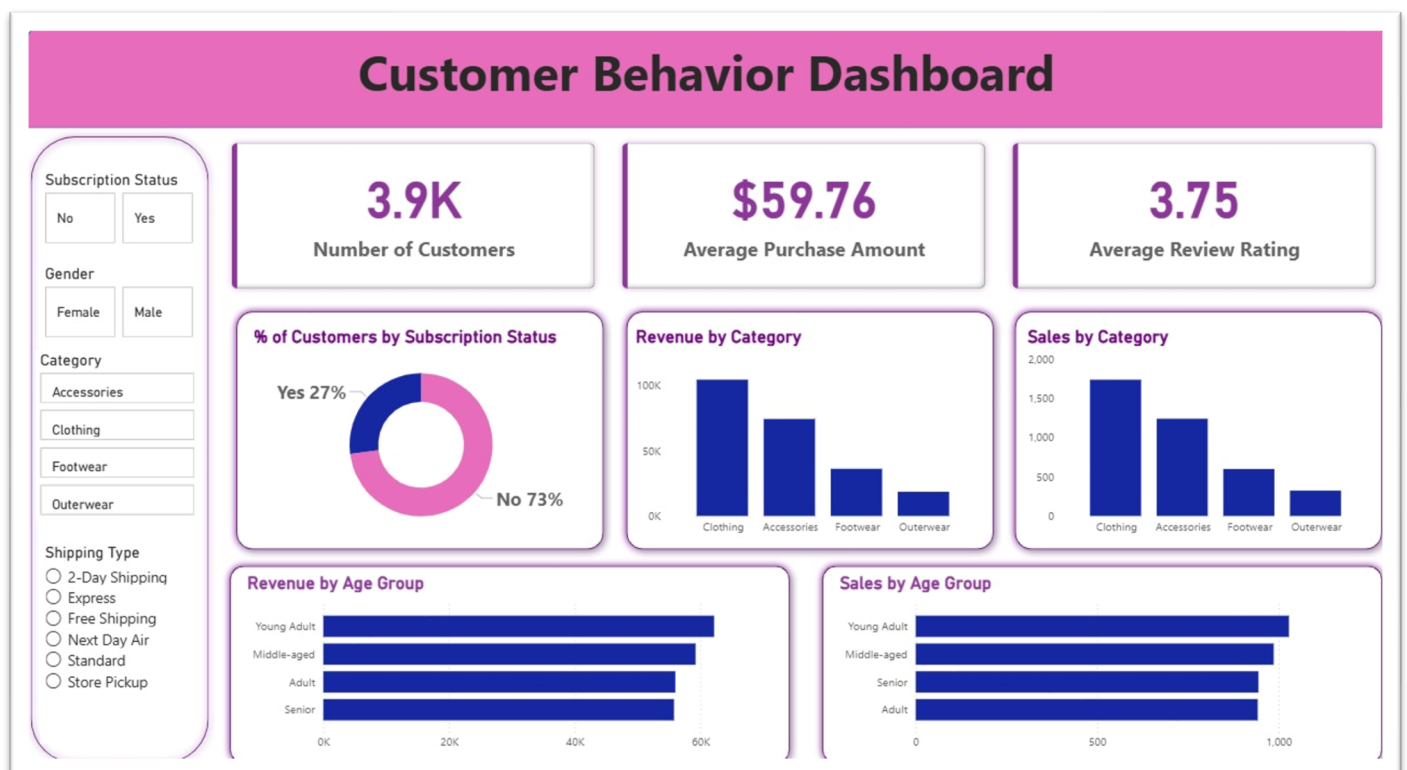
The products with the highest reported discount rates include:

- Hat
- Sneakers
- Coat
- Sweater
- Pants

Hat had the highest reported discount rate at 50%.

9. Power BI Dashboard

The Power BI dashboard converts the analytical results into an interactive visual interface.



KPI Cards

- Number of customers

- Average purchase amount
- Average review rating

Visualizations

- Subscription distribution
- Revenue by category
- Sales by category
- Revenue by age group
- Sales by age group

Filters

The dashboard includes filtering by:

- Subscription status
- Gender
- Category
- Shipping type

The supplied dashboard reports **3.9K customers**, **\$59.76 average purchase amount**, and **3.75 average rating**.

10. Business Recommendations

Recommendation 1: Improve Loyalty Programs

Develop differentiated benefits for returning and loyal customers.

Recommendation 2: Increase Subscription Adoption

Investigate why most customers remain unsubscribed and test benefits that provide recurring value.

Recommendation 3: Review Discount Strategy

High-discount products should be evaluated against sales volume and profitability before continuing aggressive discounting.

Recommendation 4: Promote High-Rated Products

Use high-rated products in product recommendations and promotional campaigns.

Recommendation 5: Segment Marketing Campaigns

Use customer behavior rather than generic demographic targeting alone.

Recommendation 6: Monitor Repeat Purchases

Track repeat-purchase frequency as a key customer-retention metric.

The supplied project similarly recommends subscription promotion, loyalty programs, discount policy review, product positioning, and targeted marketing.

11. Limitations

This project has several limitations:

1. The dataset does not provide detailed profit-margin information.
2. Customer acquisition source is unavailable.
3. Detailed web/app behavior is unavailable.
4. Long-term customer lifetime value cannot be reliably calculated from the available fields.
5. Relationships identified in the analysis should not automatically be interpreted as causal relationships.
6. The dataset represents a particular sample and may not represent all retail customers.

12. Future Scope

The project can be extended using:

Advanced Customer Segmentation

RFM analysis can classify customers using:

- Recency
- Frequency
- Monetary value

Customer Lifetime Value

Estimate expected revenue generated by customers over their relationship with the business.

Churn Prediction

Machine-learning models could identify customers who may stop purchasing.

Recommendation System

Products could be recommended based on customer purchase history.

Sales Forecasting

Historical transaction data could be used for demand forecasting if sufficient time-series information becomes available.

13. Conclusion

The Customer Shopping Behavior Analysis project demonstrates how raw transactional data can be transformed into actionable business information.

Python provides the data-cleaning and exploratory-analysis layer, SQL enables structured business analysis, and Power BI communicates the results through interactive visualizations.

The analysis highlights important areas including customer loyalty, subscription behavior, product ratings, discounts, shipping preferences, and category-level performance.

The project demonstrates the complete analytics lifecycle:

Data → Preparation → Analysis → Visualization → Insight → Business Action